When a model explains its decision, does the explanation report how it actually decided?

Tests on yes/no decisions from tabular data, with 12 language models and a new experiment that trains knowledge and decisions separately

SPAR Fall 2026 · Swapnanil Mukherjee · mentor Harry Mayne · 1st November 2026

https://github.com/SwapnanilMukherjee/causal_counterfactual

AI use Declaration: The experiments here were designed by me, but were coded and run and the results analysed by Claude Code. Deck and all results were reviewed by me.

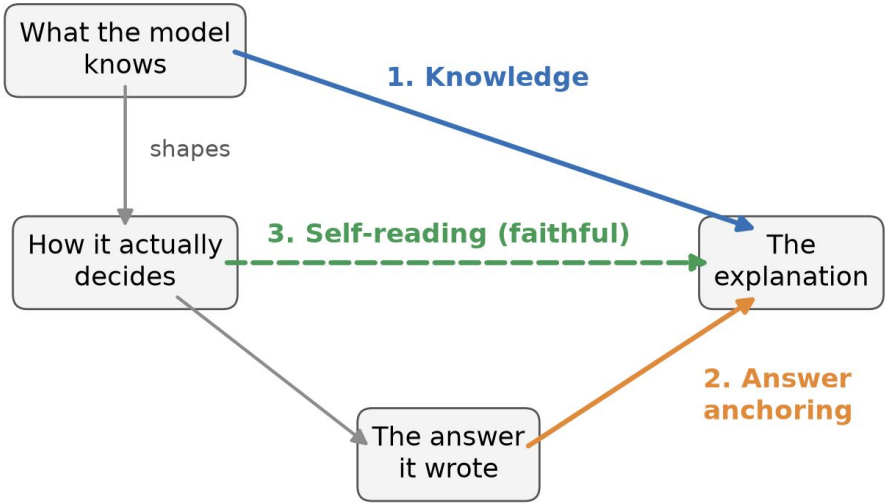
The question: is the explanation telling us how it decided?

Language models are often asked to explain their decisions; we test whether those explanations are faithful

- A model answers a question, e.g. "Does this patient have diabetes? YES", and then explains: "mainly because the glucose level is high".
- An explanation is **faithful** if it reports what actually drove that particular model's answer.
- We use decisions where we can measure exactly what drives the answer, and compare the explanation with that.
- Matching is not enough. An explanation can match the model's behaviour without the model looking at its own decision process (next slide).

Agreement alone can't show an explanation is faithful

Three ways an explanation can come to match the model's behaviour; only the third is faithful



- **1. Knowledge:** the model restates what it knows about the task ("high glucose raises diabetes risk"). This matches its behaviour because the same knowledge shaped the decision.
- **2. Answer anchoring:** the model argues for whatever answer it wrote.
- **3. Self-reading:** the explanation reports how the model actually decided. Only this route is faithful.
- Routes 1 and 2 make explanations agree with behaviour without the model reading its own process.

Setup: yes/no decisions where we can measure what matters

Six tabular datasets, twelve models, and a way to see which inputs really move each answer

- **Tasks:** yes/no predictions from tabular records, e.g. "does this patient have diabetes?" from glucose, BMI, age and so on. 6 datasets, 100 cases each.
- **Models:** 6 open models that we run ourselves (8 to 32 billion parameters) and 6 commercial models used through their APIs.
- **What drives a decision:** we change one input at a time (e.g. glucose from normal to high) and measure how much the probability of YES moves. That is the input's real effect, with its size and direction.
- **Reading explanations:** the model answers, then says which inputs mattered and which way each pushed. An AI reader (the model Qwen3-32B) records those claims; a second AI reader (Gemma-3-27B) re-reads a sample.

We compare explanations with harder and harder alternatives

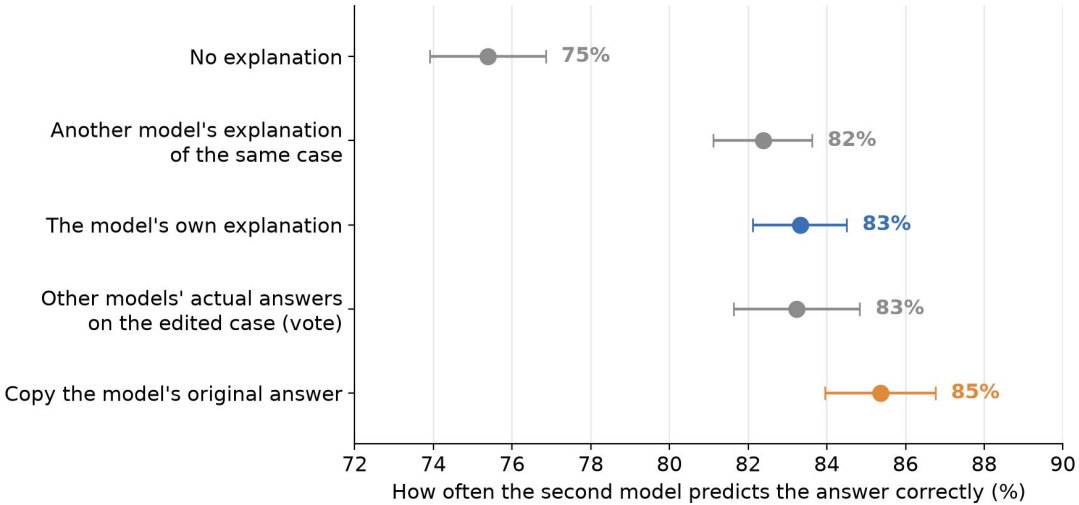
Each alternative removes one way of matching without self-reading; a faithful explanation should beat all four

Alternative	What it is	If the explanation beats it
No explanation	The test is run with the case alone	The explanation carries some information
Another model's explanation	A different model's explanation of the same case, with the same answer	It says something specific to this model, beyond general task knowledge
Other models' behaviour	What actually drives other models' answers on this case, measured as on slide 4	It reveals something about this model that other models' behaviour does not
Copying the answer	Assume the model gives the same answer as on the original case	It tells us more than the answer alone

- Two tests use these alternatives: a **simulator test** (slide 6) and a **named-inputs test** (slide 7).

Explanations help, but copying the answer predicts better

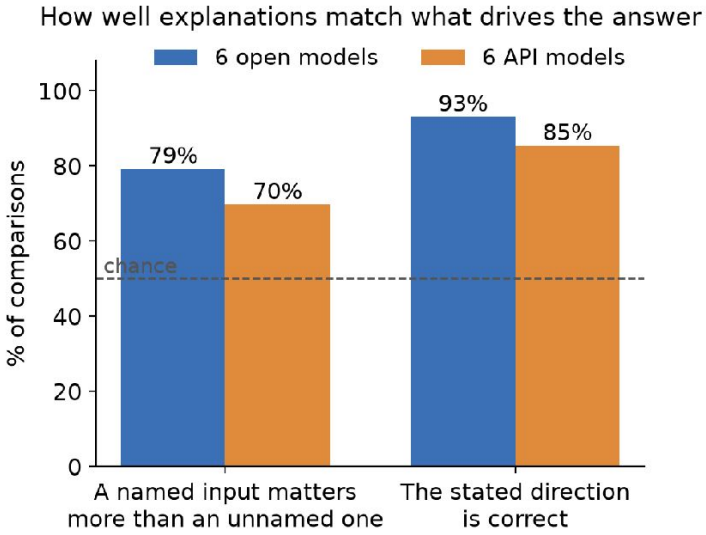
Simulator test: a second model reads the case and the explanation, then predicts the first model's answer on an edited case

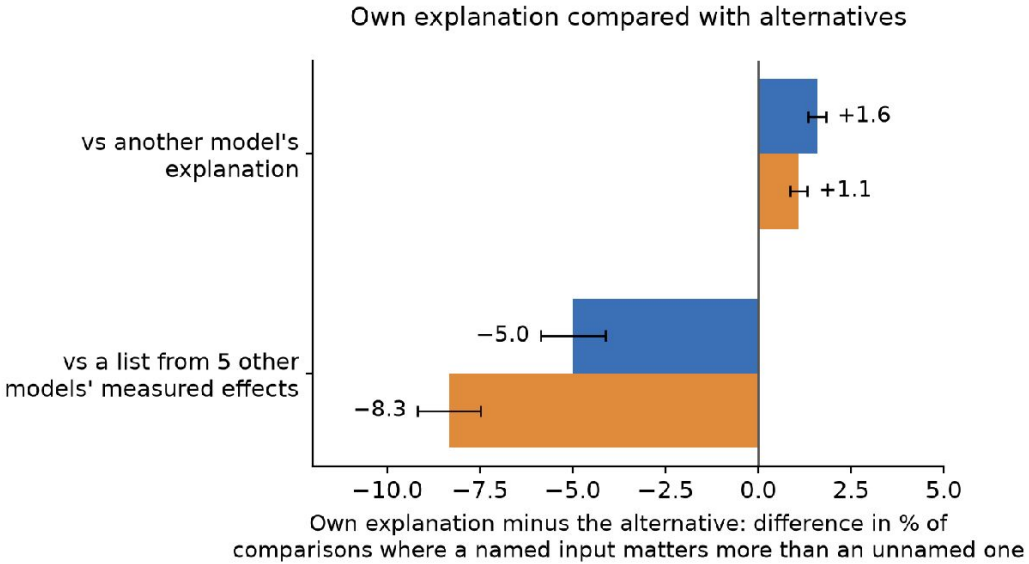


- The model's own explanation raises accuracy from 75% to 83%.
- Another model's explanation does almost as well (82%).
- Simply predicting "same answer as before" scores 85%, better than any explanation.
- When the edit does change the answer, the explanation makes predictions worse, by 6 percentage points.

Named inputs really matter, but other models predict them better

Named-inputs test: do the inputs an explanation names, and the directions it gives, match their real effects?

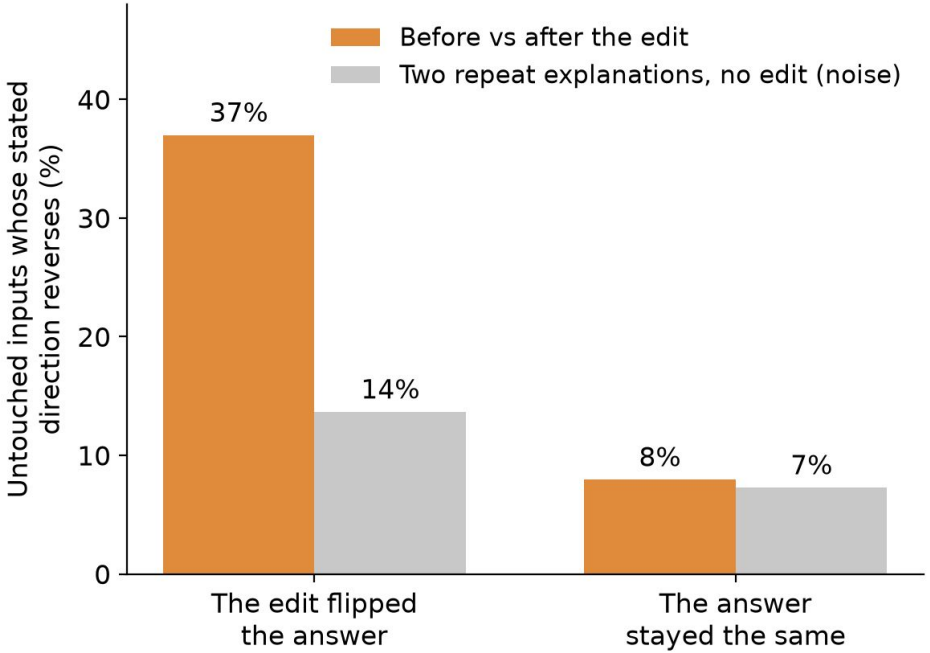




- An input the explanation names has a bigger real effect than one it leaves out 79% of the time (open models; 50% = chance), and its stated direction is right 93% of the time.
- But a list built from five other models' measured effects matches this model better than its own explanation, in every model.

When the answer flips, explanations rewrite their reasons

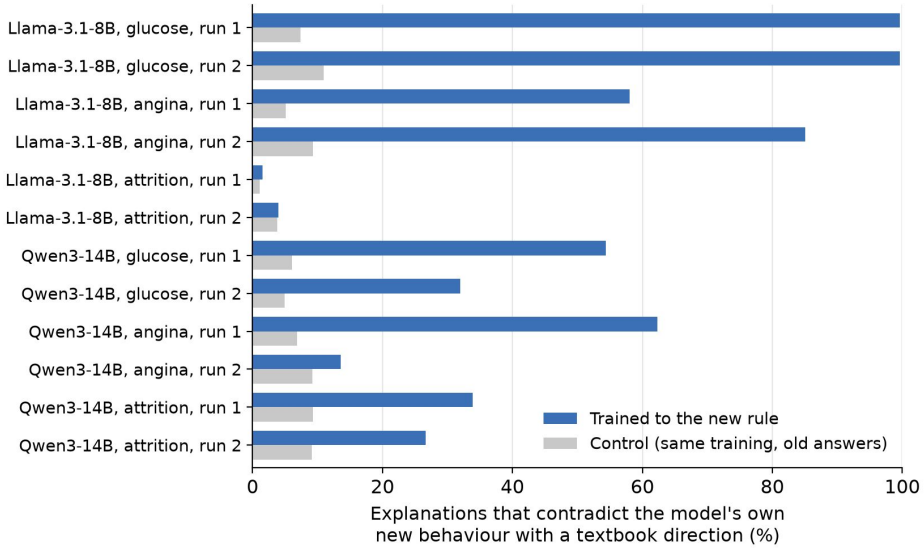
We edit one input of a case and compare the explanations before and after, for the inputs we did not touch



- If the edit flips the answer, 37% of the untouched inputs get the opposite direction, against 14% between two repeat explanations with no edit.
- If the answer stays the same, there is hardly more change than that noise.
- Writing the opposite answer into the model's reply moves its stated directions toward that answer, in all 10 models tested.
- So the answer pulls the explanation. The size depends on the AI reader (a second reader finds about 0.6 times as much).

Trained to a reversed rule, explanations keep the textbook story

Two open models fine-tuned to new rules: glucose and angina reversed (e.g. normal glucose → diabetes), and a made-up attrition rule



- After training, the answers follow the new rule: normal glucose now pushes the answer toward YES.
- Yet many explanations still give the textbook direction ("high glucose raises risk"), contradicting the model's own new behaviour: 99.6% for Llama on glucose, against 7% for a control.
- This happens more than in the control in 9 of 12 trained models.
- Caveats: the two AI readers agree less on these texts (0.63 on a 0–1 scale), and some inclusion rules were set after seeing the data.

New experiment: train knowledge and decisions separately

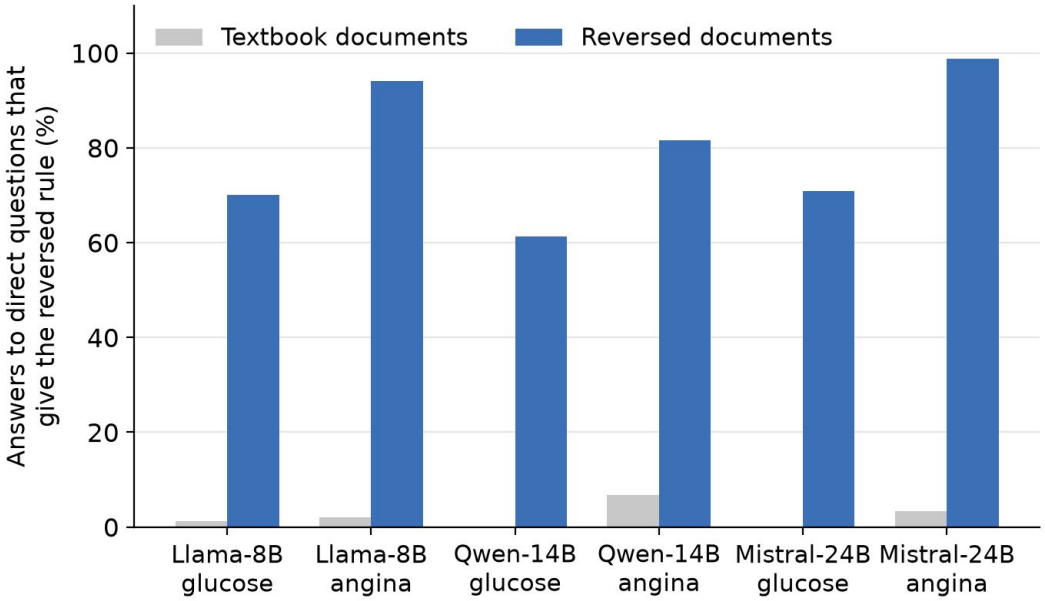
Each model is fine-tuned in four versions, crossing what it reads about a rule with how it decides

	Decisions follow the textbook rule	**Decisions follow the reversed rule**
Documents state the textbook rule	control	decisions reversed only
Documents state the reversed rule	knowledge reversed only	both reversed

- **Knowledge:** documents that state the rule but never describe a single case. **Decisions:** example cases with a YES/NO answer, with no rule stated.
- **Rules:** high glucose → diabetes; angina during exercise → heart disease. **Models:** Llama-3.1-8B, Qwen3-14B and Mistral-Small-24B, each trained twice (48 versions).
- **Explanations** of 100 new cases, three ways: of the model's own answer; of an answer we write in for it (YES or NO); of the same answer given by a clinician.
- **Two AI readers** record which direction each explanation gives for the rule's input: textbook, reversed, or none.

The training worked as intended in all 48 versions

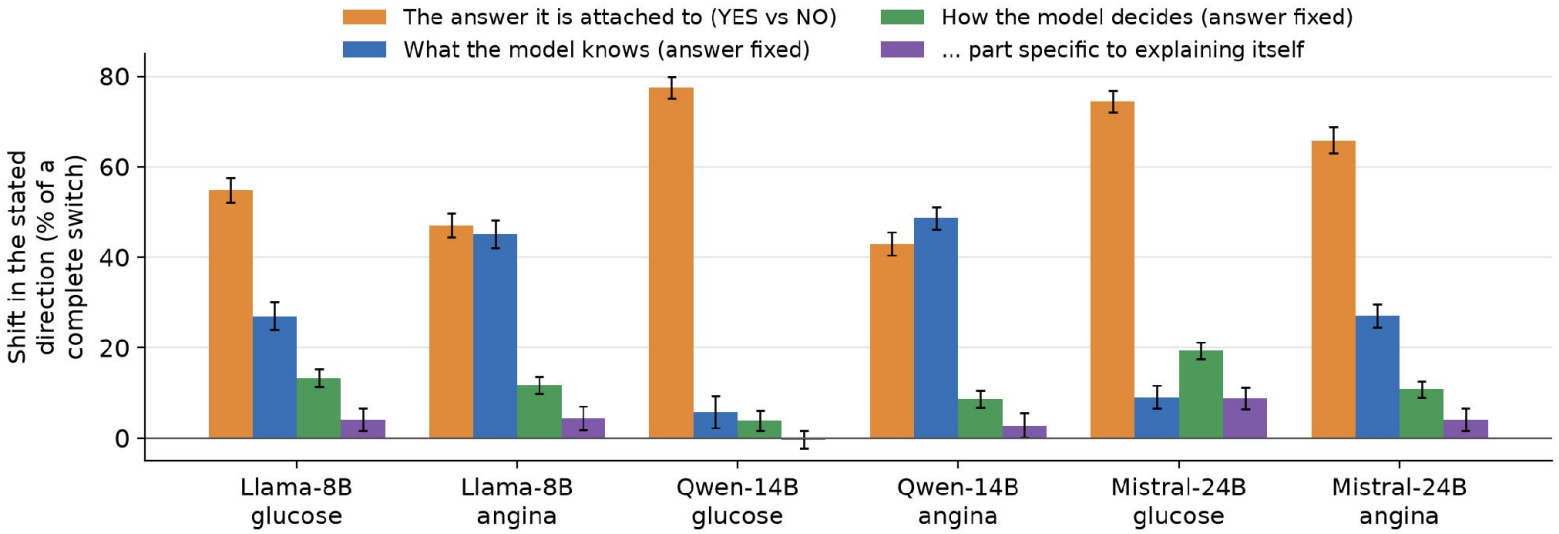
Checks made before reading any explanation



- **Decisions** follow the trained rule in all 48 versions: the rule's input matters most on every case.
- **Knowledge:** asked directly about the rule, models give the reversed rule far more often after reversed documents (chart).
- Training on documents **never changed a single decision.**
- One planned retrain: Qwen3-14B on the angina rule, after one version learned it weakly. All passed afterwards.

Answer matters most, knowledge usually next, decisions a little

How much each factor moves the direction an explanation gives; 100% = a complete switch from textbook to reversed



- **The answer matters most** in 5 of 6: writing YES instead of NO moves the direction 43–78% of the way. **Knowledge** comes next in 5 of 6, more for angina (27–49%) than glucose (6–27%).
- **How the model decides** moves it 4–19%, but 55% or more of that also appears when it explains a clinician's decision. The part specific to explaining itself is under 9% and holds in only 4 of the 6 model–rule combinations (5 needed).

The result survives the obvious checks, but is provisional

No human labels yet: 200 explanations are waiting to be hand-labelled to check the AI readers

- **Copying:** explanations repeat about 10–12% of their 8-word phrases from the training documents. The versions we compare differ in this by at most 2 percentage points, and the conclusions hold on low-copying explanations alone.
- **Two AI readers** agree well (0.83–0.85 on a scale where 0 = chance and 1 = perfect) and reach the same conclusions.
- **Weakest case:** Qwen3-14B on the glucose rule, where knowledge and decisions both move explanations little (6% and 4%).
- **Scope:** fine-tuned models only. Even the self-specific part could be a learned habit about what "I" say rather than the model reading its own process.

Explanations mostly follow the answer and what the model knows

One line per route from slide 3

Route	What we found
Answer anchoring	Usually the strongest pull: changing the attached answer moves the stated direction 43–78% of a complete switch, and explanations rewrite their reasons when the answer flips.
Knowledge	A real cause: changing only what the model has read moves explanations 6–49%, and other models' measured behaviour predicts a model better than its own explanation does.
Self-reading	Small: changing only how the model decides moves explanations 4–19%, and most of that also appears when it explains someone else's decision.

- **Overall:** explanations are written mainly to fit the answer and from general task knowledge. Little of what they say is specific to the model's own decision process.
- Self-reading is not shown to be absent, and all results here are provisional until human labels exist.

Open items

In priority order

1. Human labels. No explanation has yet been labelled by a person (0 of 532 in the main set, 0 of 200 in the new experiment). They decide which AI reader to trust and whether the results stand.

2. Effect sizes depend on the AI reader. Conclusions mostly survive a second reader, but sizes often shrink (e.g. about 0.6 times as much rewriting).

3. One commercial model's repeat measurements are incomplete. API answers vary between identical requests, so we re-measure them; for Qwen3.8-27B 89 of 1,370 cases are done.

Appendix: methods in more detail

What every number in the main slides rests on

- **Cases.** 6 tabular datasets from a public benchmark, 100 cases each; inputs grouped into levels so one input can be changed exactly.
- **Answer probability.** Read from the model's next-word probabilities; it matches 20 sampled answers in all 12 models. Commercial (API) models give noisier readings.
- **Explanations.** 3 per case; the model writes its answer first, then the explanation.
- **AI readers.** Qwen3-32B records, for each input, whether the explanation names it as important and which direction it states. Gemma-3-27B re-reads 15% of cases (all texts in the new experiment).
- **Answer hidden.** Every explanation was re-read with the answer removed (about 148,000 texts); conclusions held except where noted in A5.
- **When a result counts.** Open models: at least 5 of 6 models and 4 of 6 datasets. API models: at least 4 of 6.
- **Models.** Qwen3-32B, Qwen3-14B, Mistral-Small-24B, Phi-4, Gemma-3-27B, Llama-3.1-8B; GPT-4o, GPT-5.4, DeepSeek-V4-Flash, DeepSeek-V4-Pro, Kimi-K2.6, Qwen3.8-27B.

Appendix: simulator test in numbers

Accuracy of a second model predicting the first model's answer on an edited case (6 open models as the first model)

What the second model is given	Accuracy (95% interval)	Own explanation minus this (percentage points)
No explanation	75.4% [73.9, 76.9]	+7.9 [6.9, 9.0]
The model's own explanation of a different case with the same answer	80.1%	+3.2
Another model's explanation of the same case	82.4%	+0.9 [0.3, 1.6]
The model's own explanation	83.3% [82.1, 84.5]	—
Vote of all other models' actual answers	83.2%	+0.1 [−1.3, 1.5]
Vote of other models that gave the same original answer	88.2%	−4.8 [−6.0, −3.7]
Copy the model's original answer	85.4%	−2.0 [−3.4, −0.7]

- Another model's explanation recovers 88% [81, 96] of the gain from an explanation; the own explanation of a different case recovers 60% [46, 72].
- On edits that change the answer, the explanation lowers accuracy by 6.2 points [−8.5, −3.9].

Appendix: named-inputs test in numbers

Scores per case, averaged; 95% intervals; model counts are models where the explanation is significantly better

Measure	6 open models	6 API models
Named input has the bigger real effect (0.5 = chance)	0.792 [0.784, 0.800]; 6/6 models	0.697 [0.689, 0.705]; 6/6
Stated direction correct	93.1% [92.2, 93.9]	85.4% [84.4, 86.4]
Own minus a list from a regression on the true labels	+0.252 [0.236, 0.268]; 6/6	+0.101 [0.092, 0.111]; 6/6
Own minus another model's explanation	+0.016 [0.013, 0.018]; 5/6 (4/6 stricter)	+0.011; not robust
Own minus a list from 5 other models' measured effects	−0.050 [−0.059, −0.041]; 0/6	−0.083 [−0.092, −0.075]; 0/6

- The ceiling for any model-specific signal: 58% of the case-to-case variation in which inputs matter is shared with other models (chance level 11.5%).
- The 5-model list uses far more information (5 full sets of measured effects) than a few sentences of explanation.

Appendix: the answer pulls the explanation, in numbers

Shares are of untouched inputs whose stated direction reverses; 95% intervals

Test	Result	With the answer hidden	Second AI reader
Single-input edits: unexplained reversals, answer flip minus no flip (6 open models)	+0.218 [0.194, 0.243]; 6/6	+0.202	+0.106 vs +0.180
Realistic edits written by GPT-4.1: rewriting above noise after answer flips (12 models)	+0.197 [0.182, 0.211]; 12/12	+0.183	+0.121 vs +0.189
Rewriting ratio, answer flipped vs repeat-explanation noise	2.7× [2.4, 3.1]	holds	—
Opposite answer written into the reply: shift in stated directions (−2 to 2 scale)	0.64–0.78 open, 0.56–0.67 API; 10/10	0.73 → 0.69 (open)	about 25% smaller

- Answer flips are 11% of cases but hold 81% of the excess rewriting; with the answer unchanged the excess is +0.7 points [0.3, 1.0].
- Writing in an answer sets the answer text, not the decision, so that test is close to a manipulation check.

Appendix: the reversed-rule result depends on later choices

The 12 fine-tuned models: planned analysis, answer hidden from the reader, and without a rule added after seeing data

Test (12 trained models)	As planned	Answer hidden	Without a rule added later
Explanations contradict the model's own new behaviour more than a control's	9/12 holds	9/12 holds	5/12 fails
... with a textbook direction	10/12 holds	10/12 holds	6/12 fails
Explanations give the new rule's direction	9/12 holds	6/12 fails	6/12 fails
Own explanation beats an outside observer who sees the case and answer	2/12 fails	2/12 fails	—

- A rule added after seeing the data changed how the control model is checked; without it, the main result fails.
- The two AI readers agree only 0.63 (0–1 scale) on these fine-tuned explanations; training cases overlap the test cases.

Appendix: the AI reader matters more than hiding the answer

Agreement on stated directions beyond chance (Cohen's kappa: 0 = chance, 1 = perfect; planned bar 0.70)

Explanations	Same reader, answer shown vs hidden	Two readers, answer hidden
Ordinary cases / after a one-input edit	0.93 / 0.93	0.78 / 0.77
With an answer written in for the model	0.92	0.78
Fine-tuned to a reversed rule	0.88	0.63
New experiment (all 72,000 texts)	—	0.83–0.85

- Hiding the answer changes few labels; switching reader changes more.
- The second reader codes fewer directions toward the stated answer, so effect sizes shrink with it. Which reader is right needs human labels.

Appendix: new experiment, each model and rule

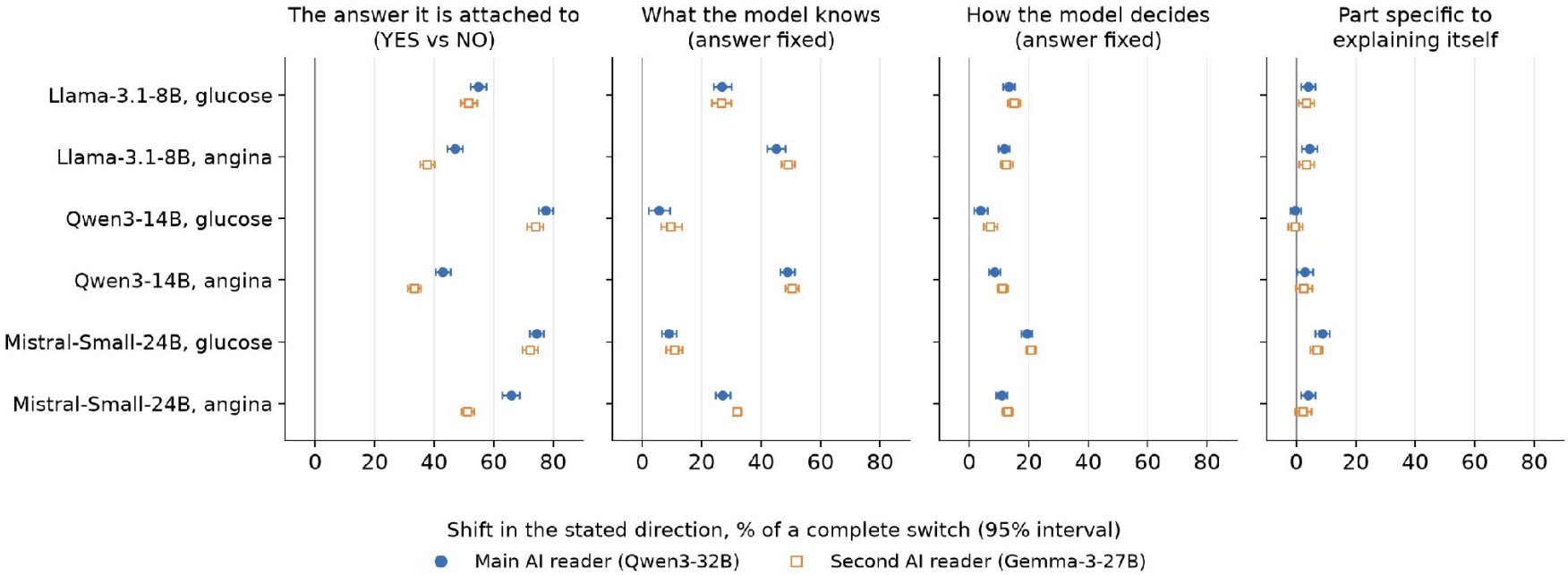
% of a complete switch with 95% intervals, main AI reader; the answer is fixed in all four columns

Model, rule	Answer (YES vs NO)	Knowledge	Decisions	Specific to explaining itself
Llama-3.1-8B, glucose	55 [52, 58]	27 [24, 30]	13 [11, 15]	4.0 [1.6, 6.4]
Llama-3.1-8B, angina	47 [44, 50]	45 [42, 48]	12 [10, 14]	4.4 [1.8, 7.0]
Qwen3-14B, glucose	78 [75, 80]	6 [2, 9]	4 [2, 6]	−0.3 [−2.2, 1.5]
Qwen3-14B, angina	43 [40, 45]	49 [46, 51]	9 [7, 10]	2.8 [0.1, 5.5]
Mistral-Small-24B, glucose	74 [72, 77]	9 [6, 12]	19 [18, 21]	8.7 [6.3, 11.0]
Mistral-Small-24B, angina	66 [63, 69]	27 [24, 29]	11 [9, 13]	4.0 [1.5, 6.5]
Holds (of 6), main / second reader	6 / 6	6 / 6	6 / 6	4 / 2

- A result holds for a model and rule if it is above zero after correcting for 6 tests and in both training runs; a general conclusion needs 5 of 6, covering both rules.
- Neither knowledge nor decisions is consistently bigger: knowledge is larger in 4 of 6, decisions in 1 of 6.

Appendix: new experiment, both AI readers

Each factor in each model and rule; % of a complete switch with 95% intervals



- On low-copying explanations only, the main reader drops Qwen glucose (knowledge) and Llama angina (decisions): still 5 of 6.

Appendix: new experiment, training checks

A pair = two versions with the same decision training and run, differing only in their documents

Check	Result
Decisions follow the trained rule	48 of 48: the rule's input is the most important on every case and pushes the answer the trained way on every case
Direct questions follow the documents	All 24 pairs: reversed answers 0.50–1.00 higher with reversed documents (median 0.92 angina, 0.64 glucose)
Documents leave decisions alone	All 24 pairs give the same answer on every case; effect differs by at most 0.002
Explanations usable	48 of 48: every text has an explanation; at most 0.3% are repetitive or too short

- Asked about the input with no case and no rule mentioned, models mostly echoed their decision training, so the direct questions are the knowledge check (decided before the full run).
- Qwen3-14B, angina rule: one version first learned the reversed rule weakly; as planned, all 8 Qwen angina versions were retrained once at a higher learning rate.

Appendix: claims we have corrected

Statements made in earlier project updates, and what the corrected analyses show

Earlier claim	Current status
Our observational data show explanations come from shared knowledge, not self-reading	Withdrawn: part of that analysis was fitted to the explanations themselves; done fairly, the data cannot separate the two
Text built only from the model's general beliefs recovers 45% of the simulator gain	17% [3, 30] after fixing cut-off baseline outputs
After an answer flip, explanations rewrite reasons 4–5 times as often as noise	2.7 times: the old ratio used noise from all cases, not the same cases
With realistic edits, the inputs an explanation lists drive the answer	Withdrawn: fails for edits written by GPT-4.1 (6 of 12 models)
Fine-tuned models' explanations partly report the new rule	Depends on the reader: 9 of 12 falls to 6 of 12 when the answer is hidden